Bismillahir Rahmanir Rahim

International Islamic University Chittagong
Department of Computer Science and Engineering
Mid-Term Examination, Autumn-2025
Course: **CSE-3521 (Computer Architecture)**
Time: **1 hour and 30 Minutes** Marks: **30**

[Answer all *three* questions; Figures in the right hand margin indicate Marks]

Q1.	a)	Define Stored-program concept. Draw and describe the computer architecture related to the Stored-program concept.	1+3	CO1 CO3
	b)	A complex statement contains the six variables **z, a, b, m, n, y**: $$z = (a + b) - (m + n) * y;$$ The variables **z, a, b, m, n and y** are assigned to the registers **St0, Ss0, Ss1, Ss2, Ss3 and Sa0** respectively. What is the compiled Assembly code?	3	CO2
	c)	What is the assembly language statement corresponding to this machine instruction below: 0xba79 ⇒ 16bits ⇒ invalid code ⇒ 32bits	3	CO2
Q2.	a)	Explain Amdahl's Law. A software team at IIUC optimizes a legacy codebase. After profiling, they discover that 50% of the program's execution time can be optimized. They manage to make this part run 2× faster, while the other 50% remains unchanged. Using Amdahl's Law, what is the overall speedup of the program?	1+4	CO1 CO2
	b)	What is the difference between CISC and RISC computers? By defining the architectures, explain how mobile and desktop chips align with CISC and RISC. Give example of the chips.	3+2	CO1
		Or		
		Explain four design principles for MIPS? Explain any three MIPS instruction formats with fields.	2+3	CO1
Q3.	a)	An old AMD A1 handles 4 requests/second. How many requests will it handle in 5 seconds? By showing the flowchart of the first method of multiplication, describe the steps.	3+2	CO2 CO3
		Or Flowc+ Table + Explanation		
		A cloud storage provider charges 12 taka per GB of storage. A company wants to buy 25 GB of storage space. Using the flow diagram of final method of multiplication, calculate the total cost.	3+2	CO2 CO3
	b)	Show the IEEE 754 binary representation of the number $-0.XY_{ten}$ in single and double precision. Indicate how you handled the normalization process. X and Y are your last two digits of student ID. (56)	4+1	CO2

1 → 4
5 → 2

International Islamic University Chittagong

Department of Computer Science & Engineering

B. Sc. in CSE Semester Mid-Term Examination, Spring 2025

Course Code: CSE 3521 Course Title: Computer Architecture

Total marks: 30 Time: 1.5 hours

		Mark	CLO	DL
1. a)	Define *CPI* and explain how it affects CPU performance. Explain with example	3	2	2
1. b)	Why is MIPS not always a reliable performance measure? Explain with example	3	2	2
1. c)	Different instruction sequences can yield different execution times for the same task. Given two compilers A and B, both targeting the same hardware:	4	2	3

Class	CPI	A Count	B Count
A	1	5	4
B	2	2	3
C	3	1	2

Assuming a 2.5 GHz clock, compute which compiler produces a faster code and justify your answer with proper reasoning.

OR Consider the following two processor designs for the same ISA:
Processor A: Clock Rate = 2.5 GHz, Average CPI = 1.6
Processor B: Clock Rate = 2.0 GHz, Average CPI = 1.2
A program takes 1.5 billion instructions to execute.
a. Compute the execution time on each processor.
b. Which processor is faster and by what percentage?
c. If performance is critical, which design would you recommend?

2. a)	Translate the following MIPS instructions into machine code: (i) addi $t1, $t2, 15 (ii) beq $s1, $s2, -8 (iii) lw $t0, 32($sp)	6	1	3

OR Translate the following machine code into MIPS instructions:
001100 01011 01010 0000000011110000
101011 10001 10000 1111111111101100
000101 00101 00110 0000000000000100

2. b)	Describe the purpose and structure of an instruction format. How it aligns with design principles.	4	1	2
3. a)	A computer system is using the IEEE 754 format to store floating-point numbers. Why is normalization important in this format? Show how the number –5.75 is normalized and represented in single precision.	6	1	3
3. b)	Why refined versions of multiplication hardware reduce its size and components? Explain with examples.	4	1	2

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International Islamic University Chittagong

Department of Computer Science & Engineering
Program: B. Sc (Engg.) in CSE Semester: 5th
Mid-Term Examination, Spring 2024
Course Code: CSE-3521 Course Title: Computer Architecture
Time: 1 hour and 30 minutes Total Marks: 30

[Parts of the same question must be answered following the serial]

			Marks	CLO	DL
Q.1.	a.	Define the terms CPI and CPU clock cycle. "MIPS can vary inversely with performance"- verify this statement with appropriate example.	4	1	1
	b.	Suppose we have two implementations of the same instruction set architecture. Machine A has a clock cycle time of 1 ns and a CPI of 3.0 for some program, and machine B has a clock cycle time of 2 ns and a CPI of 2.2 for the same program. Which machine is faster for this program, and by how much?	4	2	3

Or

Codes	Instruction Counts for instruction class(in billions)		
	A	B	c
Compiler 1	4	2	3
Compiler 2	7	6	5

Instruction Class	CPI for this Class
A	1
B	2
C	3

Assume that the computer's clock rate is 5 GHz. Which code sequence will execute faster according to MIPS? According to execution time?

	c.	*Subtract (35-7) using 2's complement.*			
Q.2.	a.	Define instructions. What is the major difference between the variables of a programming language and registers?	1+1=2	1	1
	b.	"Good Design Demands Good Compromises"-is one of the design principles of MIPS. Give proper explanations with example.	3	1	2
	c.	Translate the following MIPS Assembly Language instructions into Machine Language. You must specify the MIPS fields and Instruction Format used. (i) add $s1, $t5, $t6. (ii) bne $s2, $s4, 100 (iii) sw $s1, 100($s2)	3	2	3
	d.	What are the reasons for considering 32 bit registers each of size 32? Explain	2	1	2

Or

What is spilling register? When it is necessary to spill register?

Q.3	a.	Briefly describe different addressing modes in MIPS.	2	1	1
	b.	Show the *IEEE 754* binary representation of the number -0.75_{ten} in single and double precision.	4	2	3

Or

Describe the first version of multiplication hardware using flowchart.

	c.	Multiply $4_{ten} \times 5_{ten}$ using final version of multiplication algorithm.	4	2	3

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Mid-Term Examination, Autumn-2023

Course: **CSE-3521 (Computer Architecture)**

Time: **1hour and 30 Minutes** Marks: **30**

[Answer all *three* questions; Figures in the right hand margin indicate Marks]

Q1.	**a)**	Define Computer Architecture. Explain the subcategories of computer architecture	1+2	**CO1**
	b)	A program runs in 10 seconds on computer A, which has a 300 MHz clock. We are trying to help a computer designer build a machine, B, that will run this program in 6 seconds. The designer has determined that a substantial increase in the clock rate is possible, but this increase will affect the rest of the CPU design, causing machine B to require 1.2 times as many clock cycles as machine A for this program. What clock rate should we tell the designer to target?	4	**CO2**
	c)	Suppose we have two implementations of the same instruction set architecture. Computer1 has a CPI of 2.0 and a clock cycle time of 250 ps for some programs, and computer 2 has a CPI of 1.2 and a clock cycle time of 500 ps for same programs. Which computer is faster for this program and by how much?	3	**CO2**
Q2.	**a)**	State four design principles of MIPS	2	**CO1**
			4	**CO2**
	b)	Translate the following MIPS Assembly Language instructions into Machine L (i) sub $s2, $t5, 20 (ii) bne $s3, .$s1,100 (iii) sw $s7, 80($s2) (iv) slt $s4, $s6, $s7		

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c) What is the difference between CISC and RISC computers? 2 CO1
Or
Which registers are reserved for assembler and operating system?

d) Describe the MIPS addressing mode with figure. 2 CO1

Q3. a) Describe the first version of division Algorithm using Flow Chart. 3 CO1

Or

Describe the Final version of Multiplication Algorithm using Flow Chart.

b) Show the IEEE 754 binary representation of the number $-0.25X_{ten}$ in single and double precision 3 CO2

c) Divide $5_{ten} \div 3_{ten}$ using final version of ~~multiplication~~ division algorithm 4 CO2

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International Islamic University Chittagong

Department of Computer Science and Engineering

Mid Term Examination, Spring 2023

Course CSE-3521 (Computer Architecture)

Time 1hour and 30 Minutes Marks 30

[Answer all *three* questions; Figures in the right hand margin indicate Marks]

Q1. a) Define the following: 2 CO1

(i) Computer Architecture (ii) CPU time

b) 3 CO2

Code Form	Instruction Counts for instruction class (in billions)		
	A	B	C
Compiler 1	4	2	3
Compiler 2	10	5	Last digit of your student ID

Instruction Class	CPI for this class
A	1
B	2
C	3

Assume that the computer's clock rate is 5 GHz. Which code sequence Will execute faster according to MIPS according to execution time?

c) Suppose we have two implementations of the same instruction set architecture. Computer1 has a CPI of 3.0 and a clock cycle time of 300 ps for some programs, and computer2 has a CPI of 4.0 and a clock cycle time of 150 ps for same programs. Which computer is faster for this program and by how much? 3 CO2

d) What is CPU clock cycle and clock rate? UNIX time command is given 40.Xu 10.5s 2:50 calculate the percentage according to total elapsed time. **[N.B: X=last digit of your student ID]** 2 CO2

Q2. a) Explain different types of instruction format 2 CO1

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b) Translate the following MIPS Assembly Language instructions into Machine Language. You must specify the MIPS fields and Instruction Format used. 4 CC

(i) add $s5, $t5, 20
(ii) bne $s3, .$s1,100
(iii) sw $s7, 80($s2)
(iv) beq $s1, $s6, 24

c) What is the difference between CISC and RISC computers? 2 CO1

d) Describe the MIPS addressing mode with figure. 2 CO1

Q3. a) Describe the first version of division Algorithm using Flow Chart. 3 CO1

Or

Describe the Final version of Multiplication Algorithm using Flow Chart.

b) Show the IEEE 754 binary representation of the number $-0.375X_{ten}$ in single and double precision 3 CO2

c) Multiply $2_{ten} \times 5_{ten}$ using final version of multiplication algorithm 4 CO2

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Department of Computer Science and Engineering
B. Sc. in CSE Midterm Examination, Spring-2022
Course Code: CSE-3521 Course Title: Computer Architecture
Time: 1 hour and 30 minutes Total marks: 30

[Answer all the questions. Figures in the right margin indicate full marks.]

			CO	DL
1.				
a)	In which points of view *computer architecture* subject is very essential to study you think? Describe briefly.	4	CO1	C1
b)	Which is more significant between RISC and CISC computers in the computer technology and why?	3	CO1	C2
	OR			
b)	What do you understand about ENIAC and Abacus?	3	CO1	C2
c)	Which technology is better between LSI and VLSI you think and why?	3	CO1	C2
	OR			
c)	Define the followings: i) Computer organization, ii) Computer architecture, iii) Relative performance	3	CO1	C2
2.				
a)	Mention the *parameters* to measure the performance of a computer and describe one of those parameters.	3	CO1	C2
b)	Consider computing the overall CPI for a machine A for which the following performance measures were recorded when executing a set of benchmark programs. Assume that the clock rate of the CPU is 200 MHz:	3	CO2	C3

Instruction category	Percentage of occurrence	No. of cycles per instruction
ALU	25	2
Load & store	15	3
Branch	45	5
Others	20	2

Suppose 105 instructions will be executed. Find the overall CPI (Clock pulse per instruction) and MIPS.

c)	Suppose you have two machines X and Y. Prove that a higher MIPS machine compared to lower MIPS machine requires longer CPU time to execute the same set of benchmark programs of 2(b). Also mention your own comments in this regard.	4	CO2	C3
	OR			
c)	Translate the following instructions into machine instructions: i) add $t5, $s3, $s4 ii) beq $t1, $t2, 1000	4	CO1	C1

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3.				
a)	Mention the *registers* used in *read* operation from memory (RAM) and how?	3	CO2	C1
b)	Write the differences between Indirect, Direct addressing mode with example.	2	CO2	C2
c)	How *relative* addressing mode works? Show example.	1	CO2	C1
d)	Write a subroutine, SEARCH, to search for a value VAL in a list of N values and write the position as an output.	4	CO2	C2
	OR			
d)	Draw the hardware design of second version Multiplication Algorithm and describe using flowchart.	4	CO2	C2

END

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